

TASK 4 OUTPUTS

```
CREATE DATABASE BankDB;
```

```
USE BankDB;
```

```
CREATE TABLE Customer (
```

```
    Customer_ID INT PRIMARY KEY,
```

```
    Customer_Name VARCHAR(100) NOT NULL,
```

```
    Phone VARCHAR(15),
```

```
    Email VARCHAR(100),
```

```
    City VARCHAR(50)
```

```
);
```

```
CREATE TABLE Account (
```

```
    Account_No INT PRIMARY KEY,
```

```
    Customer_ID INT,
```

```
    Account_Type VARCHAR(20),
```

```
    Balance DECIMAL(12,2) DEFAULT 0,
```

```
    Branch VARCHAR(50),
```

```
    FOREIGN KEY (Customer_ID)
```

```
    REFERENCES Customer(Customer_ID)
```

```
);
```

```
CREATE TABLE Bank_Transaction (
```

```
    Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
```

```
    Account_No INT,
```

```
    Transaction_Type VARCHAR(20),
```

```
    Amount DECIMAL(12,2),
```

```
    Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
```

```
    FOREIGN KEY (Account_No)
```

```
    REFERENCES Account(Account_No)
```

```
);
```

```
CREATE TABLE Loan (
```

```
    Loan_ID INT PRIMARY KEY,
```

```
    Customer_ID INT,
```

```
    Loan_Type VARCHAR(30),
```

```
    Loan_Amount DECIMAL(12,2),
```

```
    Interest_Rate DECIMAL(5,2),
```

FOREIGN KEY (Customer_ID)

REFERENCES Customer(Customer_ID)

);

INSERT INTO Customer

(Customer_ID, Customer_Name, Phone, Email, City)

VALUES

(101, 'Prem Sai', '9876543210', 'premsai@gmail.com', 'Hyderabad'),

(102, 'Rahul', '9876543211', 'rahul@gmail.com', 'Vizag'),

(103, 'Arjun', '9876543212', 'arjun@gmail.com', 'Vijayawada'),

(104, 'Sai Kiran', '9876543213', 'saikiran@gmail.com', 'Hyderabad'),

(105, 'Vamsi', '9876543214', 'vamsi@gmail.com', 'Vizag'),

(106, 'Karthik', '9876543215', 'karthik@gmail.com', 'Vijayawada'),

(107, 'Rohit', '9876543216', 'rohit@gmail.com', 'Hyderabad'),

(108, 'Sanjay', '9876543217', 'sanjay@gmail.com', 'Vizag'),

(109, 'Manoj', '9876543218', 'manoj@gmail.com', 'Vijayawada'),

(110, 'Pranavi', '9876543219', 'pranavi@gmail.com', 'Hyderabad');

Result Grid					
Filter Rows:			Edit:	Export/Import:	Wrap Cell Content:
	Customer_ID	Customer_Name	Phone	Email	City
▶	101	Prem Sai	9876543210	premsai@gmail.com	Hyderabad
	102	Rahul	9876543211	rahul@gmail.com	Vizag
	103	Arjun	9876543212	arjun@gmail.com	Vijayawada
	104	Sai Kiran	9876543213	saikiran@gmail.com	Hyderabad
	105	Vamsi	9876543214	vamsi@gmail.com	Vizag
	106	Karthik	9876543215	karthik@gmail.com	Vijayawada
	107	Rohit	9876543216	rohit@gmail.com	Hyderabad
	108	Sanjay	9876543217	sanjay@gmail.com	Vizag
	109	Manoj	9876543218	manoj@gmail.com	Vijayawada
	110	Pranavi	9876543219	pranavi@gmail.com	Hyderabad
•	NOTE	NOTE	NOTE	NOTE	NOTE

INSERT INTO Account

(Account_No, Customer_ID, Account_Type, Balance, Branch)

VALUES

(10001, 101, 'Savings', 50000, 'Hyderabad'),

(10002, 102, 'Savings', 75000, 'Vizag'),

(10003, 103, 'Current', 120000, 'Vijayawada'),

(10004, 104, 'Savings', 45000, 'Hyderabad'),

(10005, 105, 'Current', 90000, 'Vizag'),

(10006, 106, 'Savings', 60000, 'Vijayawada'),

(10007, 107, 'Savings', 85000, 'Hyderabad'),

(10008, 108, 'Current', 110000, 'Vizag'),

(10009, 109, 'Savings', 55000, 'Vijayawada'),

(10010, 110, 'Savings', 70000, 'Hyderabad');

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vizag
10003	103	Current	120000.00	Vijayawada
10004	104	Savings	45000.00	Hyderabad
10005	105	Current	90000.00	Vizag
10006	106	Savings	60000.00	Vijayawada
10007	107	Savings	85000.00	Hyderabad
10008	108	Current	110000.00	Vizag
10009	109	Savings	55000.00	Vijayawada
10010	110	Savings	70000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

INSERT INTO Bank_Transaction

(Account_No, Transaction_Type, Amount)

VALUES

(10001, 'DEPOSIT', 10000),

(10002, 'DEPOSIT', 15000),

(10003, 'WITHDRAW', 20000),

(10004, 'DEPOSIT', 5000),

(10005, 'WITHDRAW', 10000),

(10006, 'DEPOSIT', 12000),

(10007, 'DEPOSIT', 8000),

(10008, 'WITHDRAW', 15000),

(10009, 'DEPOSIT', 7000),

(10010, 'DEPOSIT', 9000);

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-18 08:45:28
2	10002	DEPOSIT	15000.00	2026-08-18 08:45:28
3	10003	WITHDRAW	20000.00	2026-08-18 08:45:28
4	10004	DEPOSIT	5000.00	2026-08-18 08:45:28
5	10005	WITHDRAW	10000.00	2026-08-18 08:45:28
6	10006	DEPOSIT	12000.00	2026-08-18 08:45:28
7	10007	DEPOSIT	8000.00	2026-08-18 08:45:28
8	10008	WITHDRAW	15000.00	2026-08-18 08:45:28
9	10009	DEPOSIT	7000.00	2026-08-18 08:45:28
10	10010	DEPOSIT	9000.00	2026-08-18 08:45:28
NULL	NULL	NULL	NULL	NULL

INSERT INTO Loan

(Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate)

VALUES

(501, 101, 'Home Loan', 5000000, 7.5),

(502, 102, 'Education Loan', 1000000, 6.5),

(503, 103, 'Car Loan', 800000, 8.2),

(504, 104, 'Personal Loan', 500000, 10.5),

(505, 105, 'Home Loan', 3000000, 7.8),

(506, 106, 'Car Loan', 900000, 8.5);

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
501	101	Home Loan	5000000.00	7.50
502	102	Education Loan	1000000.00	6.50
503	103	Car Loan	800000.00	8.20
504	104	Personal Loan	500000.00	10.50
505	105	Home Loan	3000000.00	7.80
506	106	Car Loan	900000.00	8.50
NULL	NULL	NULL	NULL	NULL

SELECT * FROM Customer;

Customer_ID	Customer_Name	Phone	Email	City
101	Prem Sai	9876543210	premsai@gmail.com	Hyderabad
102	Rahul	9876543211	rahul@gmail.com	Vizag
103	Arjun	9876543212	arjun@gmail.com	Vijayawada
104	Sai Kiran	9876543213	saikiran@gmail.com	Hyderabad
105	Vamsi	9876543214	vamsi@gmail.com	Vizag
106	Karthik	9876543215	karthik@gmail.com	Vijayawada
107	Rohit	9876543216	rohit@gmail.com	Hyderabad
108	Sanjay	9876543217	sanjay@gmail.com	Vizag
109	Manoj	9876543218	manoj@gmail.com	Vijayawada
110	Pranavi	9876543219	pranavi@gmail.com	Hyderabad
NULL	NULL	NULL	NULL	NULL

SELECT * FROM Account;

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vizag
10003	103	Current	120000.00	Vijayawada
10004	104	Savings	45000.00	Hyderabad
10005	105	Current	90000.00	Vizag
10006	106	Savings	60000.00	Vijayawada
10007	107	Savings	85000.00	Hyderabad
10008	108	Current	110000.00	Vizag
10009	109	Savings	55000.00	Vijayawada
10010	110	Savings	70000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

SELECT * FROM Bank_Transaction;

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-18 08:45:28
2	10002	DEPOSIT	15000.00	2026-08-18 08:45:28
3	10003	WITHDRAW	20000.00	2026-08-18 08:45:28
4	10004	DEPOSIT	5000.00	2026-08-18 08:45:28
5	10005	WITHDRAW	10000.00	2026-08-18 08:45:28
6	10006	DEPOSIT	12000.00	2026-08-18 08:45:28
7	10007	DEPOSIT	8000.00	2026-08-18 08:45:28
8	10008	WITHDRAW	15000.00	2026-08-18 08:45:28
9	10009	DEPOSIT	7000.00	2026-08-18 08:45:28
10	10010	DEPOSIT	9000.00	2026-08-18 08:45:28
NULL	NULL	NULL	NULL	NULL

SELECT * FROM Loan;

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
501	101	Home Loan	5000000.00	7.50
502	102	Education Loan	1000000.00	6.50
503	103	Car Loan	800000.00	8.20
504	104	Personal Loan	500000.00	10.50
505	105	Home Loan	3000000.00	7.80
506	106	Car Loan	900000.00	8.50
NULL	NULL	NULL	NULL	NULL

DELIMITER //

```
CREATE PROCEDURE GetAllCustomers()
```

```
BEGIN
```

```
    SELECT * FROM Customer;
```

```
END //
```

```
DELIMITER ;
```

```
CALL GetAllCustomers();
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Customer_ID	Customer_Name	Phone	Email	City
101	Prem Sai	9876543210	premsai@gmail.com	Hyderabad
102	Rahul	9876543211	rahul@gmail.com	Vizag
103	Arjun	9876543212	arjun@gmail.com	Vijayawada
104	Sai Kiran	9876543213	saikiran@gmail.com	Hyderabad
105	Vamsi	9876543214	vamsi@gmail.com	Vizag
106	Karthik	9876543215	karthik@gmail.com	Vijayawada
107	Rohit	9876543216	rohit@gmail.com	Hyderabad
108	Sanjay	9876543217	sanjay@gmail.com	Vizag
109	Manoj	9876543218	manoj@gmail.com	Vijayawada
110	Pranavi	9876543219	pranavi@gmail.com	Hyderabad

```
DELIMITER //
```

```
CREATE PROCEDURE GetAccountDetails(
```

```
    IN p_Account_No INT
```

```
)
```

```
BEGIN
```

```
    SELECT *
```

```
    FROM Account
```

```
    WHERE Account_No = p_Account_No;
```

```
END //
```

```
DELIMITER ;
```

```
CALL GetAccountDetails(10001);
```

Result Grid  Filter Rows: | Export:  | Wrap Cell Content: 

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad

```
DELIMITER //
```

```
CREATE PROCEDURE GetCustomerAccounts(
```

```
    IN p_Customer_ID INT
```

```
)
```

```
BEGIN
```

```
    SELECT
```

```
        C.Customer_ID,
```

```
        C.Customer_Name,
```

```
        A.Account_No,
```

```

        A.Account_Type,
        A.Balance,
        A.Branch
    FROM Customer C
    JOIN Account A
    ON C.Customer_ID = A.Customer_ID
    WHERE C.Customer_ID = p_Customer_ID;
END //

DELIMITER ;

CALL GetCustomerAccounts(101);

```

Result Grid						
Filter Rows:		Export:		Wrap Cell Content:		
Customer_ID	Customer_Name	Account_No	Account_Type	Balance	Branch	
101	Prem Sai	10001	Savings	50000.00	Hyderabad	

```

DELIMITER //

CREATE PROCEDURE DepositMoney(
    IN p_Account_No INT,
    IN p_Amount DECIMAL(12,2)
)
BEGIN
    UPDATE Account
    SET Balance = Balance + p_Amount
    WHERE Account_No = p_Account_No;
END //

DELIMITER ;

CALL DepositMoney(10001, 5000);

```

```

DELIMITER //

CREATE PROCEDURE WithdrawMoney(
    IN p_Account_No INT,
    IN p_Amount DECIMAL(12,2)
)
BEGIN
    UPDATE Account
    SET Balance = Balance - p_Amount

```

```
WHERE Account_No = p_Account_No;  
END //
```

```
DELIMITER ;
```

```
CALL WithdrawMoney(10001, 3000);
```

```
DELIMITER //
```

```
CREATE TRIGGER CheckBalance  
BEFORE UPDATE ON Account  
FOR EACH ROW  
BEGIN  
    IF NEW.Balance < 0 THEN  
        SIGNAL SQLSTATE '45000'  
        SET MESSAGE_TEXT = 'Transaction failed: Insufficient balance';  
    END IF;  
END //
```

```
DELIMITER ;
```

```
DELIMITER //
```

```
CREATE TRIGGER CheckTransactionAmount  
BEFORE INSERT ON Bank_Transaction  
FOR EACH ROW  
BEGIN  
    IF NEW.Amount <= 0 THEN  
        SIGNAL SQLSTATE '45000'  
        SET MESSAGE_TEXT = 'Transaction amount must be greater than zero';  
    END IF;  
END //
```

```
DELIMITER ;
```

```
CREATE TABLE Transaction_Audit (  
    Audit_ID INT PRIMARY KEY AUTO_INCREMENT,  
    Transaction_ID INT,  
    Account_No INT,  
    Transaction_Type VARCHAR(20),  
    Amount DECIMAL(12,2),  
    Audit_Date DATETIME DEFAULT CURRENT_TIMESTAMP  
);
```

```
DELIMITER //
```

```
CREATE TRIGGER TransactionAudit  
AFTER INSERT ON Bank_Transaction  
FOR EACH ROW  
BEGIN
```

```
    INSERT INTO Transaction_Audit
```

```
    (
```

```
        Transaction_ID,
```

```
        Account_No,
```

```
        Transaction_Type,
```

```
        Amount
```

```
    )
```

```
VALUES
```

```
    (
```

```
        NEW.Transaction_ID,
```

```
        NEW.Account_No,
```

```
        NEW.Transaction_Type,
```

```
        NEW.Amount
```

```
    );
```

```
END //
```

```
DELIMITER ;
```

```
DELIMITER //
```

```
CREATE TRIGGER UpdateBalanceAfterTransaction
AFTER INSERT ON Bank_Transaction
FOR EACH ROW
BEGIN
    IF NEW.Transaction_Type = 'DEPOSIT' THEN
        UPDATE Account
        SET Balance = Balance + NEW.Amount
        WHERE Account_No = NEW.Account_No;
    ELSEIF NEW.Transaction_Type = 'WITHDRAW' THEN
        UPDATE Account
        SET Balance = Balance - NEW.Amount
        WHERE Account_No = NEW.Account_No;
    END IF;
END //
```

DELIMITER ;

DELIMITER //

```
CREATE TRIGGER PreventInsufficientWithdrawal
BEFORE INSERT ON Bank_Transaction
FOR EACH ROW
BEGIN
    DECLARE CurrentBalance DECIMAL(12,2);

    SELECT Balance
    INTO CurrentBalance
    FROM Account
    WHERE Account_No = NEW.Account_No;

    IF NEW.Transaction_Type = 'WITHDRAW'
    AND NEW.Amount > CurrentBalance THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Withdrawal failed: Insufficient balance';
    END IF;
```

END //

DELIMITER ;

DELIMITER //

CREATE PROCEDURE TransferMoney(

 IN SenderAccount INT,

 IN ReceiverAccount INT,

 IN TransferAmount DECIMAL(12,2)

)

BEGIN

 DECLARE SenderBalance DECIMAL(12,2);

 SELECT Balance

 INTO SenderBalance

 FROM Account

 WHERE Account_No = SenderAccount;

 IF SenderBalance < TransferAmount THEN

 SIGNAL SQLSTATE '45000'

 SET MESSAGE_TEXT = 'Transfer failed: Insufficient balance';

 ELSE

 UPDATE Account

 SET Balance = Balance - TransferAmount

 WHERE Account_No = SenderAccount;

 UPDATE Account

 SET Balance = Balance + TransferAmount

 WHERE Account_No = ReceiverAccount;

 END IF;

END //

DELIMITER ;

```
CALL TransferMoney(10001, 10002, 5000);
```

```
SELECT *
```

```
FROM Account
```

```
WHERE Account_No IN (10001,10002);
```

```
DELIMITER //
```

```
CREATE PROCEDURE GetCustomerLoans(
```

```
    IN p_Customer_ID INT
```

```
)
```

```
BEGIN
```

```
    SELECT
```

```
        C.Customer_Name,
```

```
        L.Loan_ID,
```

```
        L.Loan_Type,
```

```
        L.Loan_Amount,
```

```
        L.Interest_Rate
```

```
FROM Customer C
```

```
JOIN Loan L
```

```
ON C.Customer_ID = L.Customer_ID
```

```
WHERE C.Customer_ID = p_Customer_ID;
```

```
END //
```

```
DELIMITER ;
```

```
CALL GetCustomerLoans(101);
```

```
DELIMITER //
```

```
CREATE PROCEDURE HighBalanceAccounts(
```

```
    IN MinimumBalance DECIMAL(12,2)
```

```
)
```

```
BEGIN
```

```
    SELECT *
```

```
FROM Account

WHERE Balance >= MinimumBalance

ORDER BY Balance DESC;

END //


DELIMITER ;


CALL HighBalanceAccounts(50000);


DELIMITER //


CREATE PROCEDURE GetBalance(
    IN p_Account_No INT,
    OUT p_Balance DECIMAL(12,2)
)
BEGIN
    SELECT Balance
    INTO p_Balance
    FROM Account
    WHERE Account_No = p_Account_No;
END //


DELIMITER ;


CALL GetBalance(10001, @CurrentBalance);


SELECT @CurrentBalance;


SELECT *
FROM Transaction_Audit;


SHOW TRIGGERS;


SHOW CREATE TRIGGER CheckBalance;
```

```
SHOW PROCEDURE STATUS
```

```
WHERE Db = 'BankDB';
```

```
DELIMITER //
```

```
CREATE PROCEDURE GetBranchAccounts(
```

```
    IN p_Branch VARCHAR(50)
```

```
)
```

```
BEGIN
```

```
    SELECT *
```

```
    FROM Account
```

```
    WHERE Branch = p_Branch;
```

```
END //
```

```
DELIMITER ;
```

```
CALL GetBranchAccounts('Hyderabad');
```

```
DELIMITER //
```

```
CREATE PROCEDURE GetCustomerFullDetails(
```

```
    IN p_Customer_ID INT
```

```
)
```

```
BEGIN
```

```
    SELECT
```

```
        C.Customer_Name,
```

```
        C.Phone,
```

```
        C.Email,
```

```
        A.Account_No,
```

```
        A.Account_Type,
```

```
        A.Balance
```

```
    FROM Customer C
```

```
    JOIN Account A
```

```
    ON C.Customer_ID = A.Customer_ID
```

```
    WHERE C.Customer_ID = p_Customer_ID;
```

END //

DELIMITER ;

CALL GetCustomerFullDetails(101);

DELIMITER //

CREATE PROCEDURE GetCustomerTotalBalance(

IN p_Customer_ID INT

)

BEGIN

SELECT

C.Customer_Name,

SUM(A.Balance) AS Total_Balance

FROM Customer C

JOIN Account A

ON C.Customer_ID = A.Customer_ID

WHERE C.Customer_ID = p_Customer_ID

GROUP BY C.Customer_Name;

END //

DELIMITER ;

CALL GetCustomerTotalBalance(101);

DELIMITER //

CREATE PROCEDURE GetTopFiveAccounts()

BEGIN

SELECT *

FROM Account

ORDER BY Balance DESC

LIMIT 5;

END //

DELIMITER ;

CALL GetTopFiveAccounts();

DELIMITER //

CREATE PROCEDURE CountAccountsByBranch()

BEGIN

SELECT

Branch,

COUNT(*) AS Number_of_Accounts

FROM Account

GROUP BY Branch;

END //

DELIMITER ;

CALL CountAccountsByBranch();

DELIMITER //

CREATE PROCEDURE GetCustomersWithLargeLoans(

IN MinimumLoan DECIMAL(12,2)

)

BEGIN

SELECT

C.Customer_Name,

L.Loan_ID,

L.Loan_Type,

L.Loan_Amount

FROM Customer C

JOIN Loan L

ON C.Customer_ID = L.Customer_ID

WHERE L.Loan_Amount > MinimumLoan;

END //

DELIMITER ;

CALL GetCustomersWithLargeLoans(1000000);

DELIMITER //

CREATE PROCEDURE GetTransactionHistory(

IN p_Account_No INT

)

BEGIN

SELECT *

FROM Bank_Transaction

WHERE Account_No = p_Account_No

ORDER BY Transaction_Date DESC;

END //

DELIMITER ;

CALL GetTransactionHistory(10001);

DELIMITER //

CREATE PROCEDURE GetAllAccountSummary()

BEGIN

SELECT

A.Account_No,

C.Customer_Name,

A.Account_Type,

A.Balance,

A.Branch

FROM Account A

JOIN Customer C

ON A.Customer_ID = C.Customer_ID;

END //

DELIMITER ;

CALL GetAllAccountSummary();

DELIMITER //

CREATE PROCEDURE CalculateLoanInterest(

IN p_Loan_Amount DECIMAL(12,2),

IN p_Interest_Rate DECIMAL(5,2),

IN p_Years INT

)

BEGIN

SELECT

(p_Loan_Amount * p_Interest_Rate * p_Years) / 100

AS Simple_Interest,

p_Loan_Amount +

((p_Loan_Amount * p_Interest_Rate * p_Years) / 100)

AS Total_Amount;

END //

DELIMITER ;

CALL CalculateLoanInterest(1000000, 7.5, 5);